
GENERAL RELATIVITY

Tutorial problem set 5, 11.10.2019.

■ PROBLEM 13 Causality in GR.

Consider the following three 3-dimensional space-time metrics,

$$(i) \quad ds^2 = -c^2 dt^2 + dr^2 + r^2 d\varphi^2, \quad (13.1)$$

$$(ii) \quad ds^2 = -c^2 dt^2 + \frac{d\rho^2}{1 - \kappa\rho^2} + \rho^2 d\varphi^2, \quad (\kappa > 0 \text{ or } \kappa < 0) \quad (13.2)$$

$$(iii) \quad ds^2 = -c^2 dt^2 + R^2 d\vartheta^2 + R^2 \sin^2(\vartheta) d\varphi^2. \quad (R > 0) \quad (13.3)$$

The ranges of coordinates are $\varphi \in [0, 2\pi)$, $r \in [0, \infty)$, $\vartheta \in [0, \pi)$, $\rho \in [-\frac{1}{\sqrt{\kappa}}, \frac{1}{\sqrt{\kappa}}]$ for $\kappa > 0$ where $\rho = \frac{1}{\sqrt{\kappa}}$ and $\rho = -\frac{1}{\sqrt{\kappa}}$ are identified, $\rho \in [0, \infty]$ for $\kappa < 0$.

- (a) Can you recognize what are cases (ii) for $\kappa > 0$ and cases (iii)? Can you parametrize a sphere in cylindrical coordinates?
- (b) Derive what are the curves traced by light in these three spaces in given coordinates and sketch the light cones. Note that there is a rotation symmetry in all three metrics with respect to angle φ .
- (c) Sketch the closed achronal surface S and the future and past domains of dependence $D^+(S)$ and $D^-(S)$, if for each of the cases S is defined to be

$$(i) \quad r \in [0, R], \varphi \in [0, 2\pi); \quad (13.4)$$

$$(ii) \quad \kappa > 0: \quad \rho \in [0, \rho_0], \varphi \in [0, 2\pi) \quad (0 < \rho_0 < \frac{1}{\sqrt{\kappa}}) \quad (13.5)$$

$$\kappa < 0: \quad \rho \in [0, \rho_0], \varphi \in [0, 2\pi); \quad (13.6)$$

$$(iii) \quad \vartheta \in [0, \vartheta_0], \varphi \in [0, 2\pi) \quad (0 < \vartheta_0 < \pi). \quad (13.7)$$

- (d) Sketch the Cauchy surface for all the cases.

■ PROBLEM 14 Integration measure in terms of the ϵ -tensor

Recall from last week that the ϵ -tensor is given by

$$\epsilon \equiv \sqrt{|g|} \frac{1}{n!} \tilde{\epsilon}_{\mu_1 \dots \mu_n} dx^{\mu_1} \wedge dx^{\mu_2} \wedge \dots \wedge dx^{\mu_n} \equiv \sqrt{|g|} d^n x \quad (14.1)$$

The volume of an n -manifold M is defined as the integral

$$V[M] \equiv \int_M \epsilon. \quad (14.2)$$

For a scalar field $\phi : M \rightarrow \mathbb{R}$, the integral over M is then defined as

$$I_M[\phi] \equiv \int_M \epsilon \phi. \quad (14.3)$$

- (a) Consider the unit 2-sphere S^2 . Calculate ϵ and the 'volume' $V[S^2]$.
Furthermore, given $\phi, \psi : S^2 \rightarrow \mathbb{R}$, $\phi(x, y, z) = x$ and $\psi(x, y, z) = x^2$, calculate the integrals $I_{S^2}[\phi]$ and $I_{S^2}[\psi]$.
- (b) Now consider the torus T^2 , with radii r and R ($r < R$). Calculate $V[T^2]$ and the integrals $I_{T^2}[\phi]$ and $I_{T^2}[\psi]$.
- (c) Find the area of a cone, with deficit angle $2\pi\Delta$ and length L ?
Hint: Show first that the metric can be written as, $ds^2 = d\rho^2 + (1 - \Delta)\rho^2 d\phi^2$, ($0 \leq \rho \leq L, 0 \leq \phi < 2\pi$), where $0 \leq \Delta < 1$ is a constant. What is the physical meaning of Δ ?

■ **PROBLEM 15** Christoffel symbols and covariant derivatives I.

In the lecture you it was shown that Christoffel symbols compatible with the metric have the following form

$$\Gamma_{\mu\nu}^{\alpha} = \frac{1}{2}g^{\alpha\beta} \left(\partial_{\mu}g_{\nu\beta} + \partial_{\nu}g_{\mu\beta} - \partial_{\beta}g_{\mu\nu} \right). \quad (15.1)$$

- (a) Show that Christoffel symbols are not tensors, i.e. show that under coordinate transformation $x \rightarrow \tilde{x}$ they transform in the following manner

$$\tilde{\Gamma}_{\mu\nu}^{\alpha} = \frac{\partial \tilde{x}^{\alpha}}{\partial x^{\beta}} \frac{\partial x^{\rho}}{\partial \tilde{x}^{\mu}} \frac{\partial x^{\sigma}}{\partial \tilde{x}^{\nu}} \Gamma_{\rho\sigma}^{\beta} + \frac{\partial \tilde{x}^{\alpha}}{\partial x^{\beta}} \frac{\partial^2 x^{\beta}}{\partial \tilde{x}^{\mu} \partial \tilde{x}^{\nu}}. \quad (15.2)$$

(Note that this is the reason why we are not careful about index placement with Christoffel symbols). Are they tensors under linear coordinate transformations?

- (b) Using the result of problem (a) and the definition of the covariant derivative, show explicitly that the following two quantities transform as tensors

$$(i) \quad \nabla_{\mu}\omega_{\nu}, \quad (ii) \quad \nabla_{\mu}V^{\nu}, \quad (15.3)$$

and that $\nabla_{\mu}V^{\mu}$ is a scalar quantity.